LESSON PLAN

AUXILIARY EQUIPMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Operation & functions of bearing
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Operation & functions of bearing.
5. Specific Objectives	18.1.3.1 Trainee will be able to explain operation & functions of bearing

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of bearing operation using cutaway model; Safety: Handle rotating parts with care; Supervision: Instructor monitors trainee interaction with model
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a bearing in an aircraft engine?
(b) Name two types of friction that bearings reduce.
(c) Why are bearings essential for rotating shafts in auxiliary equipment?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic principle of bearing • Function of reducing friction and wear • Role in supporting radial and axial loads • Importance of lubrication in bearing operation • Basic construction of rolling element bearing	L & Ds	PPT, IB	15 min
(b)	• Detailed operation of rolling element bearings • Mechanism of load distribution in races • Effect of speed and load on bearing life • Common failure modes due to improper operation • Maintenance requirements for optimal function	L & Ds	PPT, PDF	20 min
(c)	• Summary of key operational principles • Review of functions and load types • Importance of correct installation practices • Connection to auxiliary equipment systems • Preparation for next lesson on classification	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a bearing reduces friction in a rotating shaft.
(b) Describe the function of bearing races in load support.
(c) List three factors affecting bearing operation and function.

12. Home Assignment and Follow Up	Prepare a brief note on how the operation of bearings relates to the classification of bearings covered in LP 08.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Classification of bearing	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min